Yifei Dong

Lindstedtsvägen 24, 114 28 Stockholm
Sweden
Website: yifeidong0.github.io/
Email: yifeid@kth.se

Education

2025 **Duke University**, *Visiting Scholar*, Robot Dexterity Lab, with Prof. Xianyi Cheng

2023 - Now **KTH Royal Institute of Technology**, *PhD Candidate*, Robotics, Computer Science

2019 - 2022 **ETH Zurich**, *M.Sc.*, Mechanical Engineering

2015 - 2019 **Shanghai Jiao Tong University**, *B.Sc.*, Mechanical Engineering
Tsien Hsue-shen Program & Zhiyuan Honors Program of Engineering

Employment

2023 - Now **Robotics, Perception and Learning, KTH**, *Stockholm*
Doctoral Student, with Assoc. Prof. Florian Pokorny

2022 **Idiap, EPFL**, *Martigny*, *Research Assistant*, with Dr. Sylvain Calinon
Phase estimation on Riemannian manifold. Code contribution: Robotics Code From Scratch

2022 **F&P Robotics**, *Zürich*, *Software Engineer Intern*
Motion planning of door opening and human-robot interaction for a mobile service robot.

2018 **NIO Inc.**, *Shanghai*, *Software Engineer Intern*
LiDAR simulation of an autonomous vehicle in various traffic scenes.

Publications

Journals and Conference Proceedings (* indicates equal contribution):

[J1] Yifei Dong, Xianyi Cheng, Florian Pokorny, *Characterizing Manipulation Robustness through Energy Margin and Caging Analysis*, **RA-L**, 2024, *Paper*

[J2] Haofei Lu, Yifei Dong, Zehang Weng, Florian Pokorny, Jens Lundell, Danica Kragic, *Grasping a Handful: Sequential Multi-Object Dexterous Grasp Generation*, **RA-L**, 2025, *Paper* | *Website*

[C1] Kei Ikemura*, Yifei Dong*, Florian Pokorny, *Latent Diffeomorphic Co-Design of End-Effectors for Deformable and Fragile Object Manipulation*, Proceedings of Robotics: Science and Systems (**RSS**), 2026, *Paper*

[C2] Yifei Dong*, Yan Zhang*, Sylvain Calinon, Florian Pokorny, *Robustness-Aware Tool Selection and Manipulation Planning with Learned Energy-Informed Guidance*, **ICRA**, 2026, *Paper*

[C3] Yifei Dong*, Shaohang Han* et al., *CageCoOpt: Enhancing Manipulation Robustness through Caging-Guided Morphology and Policy Co-Optimization*, **IROS**, 2025, *Paper*

[C4] Yifei Dong, Florian Pokorny, *Quasi-static Soft Fixture Analysis of Rigid and Deformable Objects*, **ICRA**, 2024, *Paper*

[C5] David Cáceres Domínguez et al., *The First WARA Robotics Mobile Manipulation Challenge–Lessons Learned*, European Conference on Mobile Robots, 2025, *Paper* | *Video*

[C6] Yue Chen et al., *DUET: Joint Exploration of User Item Profiles in Recommendation System*, ACL findings, 2026, *Paper*

Workshop Paper, Under Review and Others:

[1] Kei Ikemura, Yifei Dong et al. *Efficient End-effector Co-Design by Demonstration for Deformable Fragile Object Manipulation*, RSS Workshop on Robot Hardware-Aware Intelligence (Spotlight), 2025, *Paper*

[2] David Blanco-Mulero, Yifei Dong, Julia Borras, Florian Pokorny and Carme Torras, *T-DOM: A Taxonomy for Robotic Manipulation of Deformable Objects*, tech report, 2025, *Preprint* | *Website*

[3] Kei Ikemura*, Yifei Dong* et al., *Sim-to-Real Gentle Manipulation of Deformable and Fragile Objects with Stress-Guided Reinforcement Learning*, tech report, 2025, *Preprint*

[4] Christoffer K. Øhrstrøm, Rafael I. C. Muchacho, Yifei Dong et al., *Where to Attend: A Principled Vision-Centric Position Encoding with Parabolas*, tech report, 2026, *Preprint*

[5] Yue Chen et al., *WarriorMath: Empowering Mathematical Reasoning for Large Language Models via Expert Battles*, tech report, 2025, *Preprint*

[6] Yifei Dong, Zhanyi Sun, Lujie Yang, Manuel Baum, Shuran Song, Florian Pokorny, Xianyi Cheng, *Robustness of Robotic Manipulation: Foundations and Frontiers*, survey, 2026,

[7] Yan Zhang, Yiming Li, Yifei Dong, Florian Pokorny, Sylvain Calinon, *Physics-Informed Eikonal Caging for Whole-Arm Manipulation Planning*, tech report, 2026,

Academic Activities

Talk, Academic Service, Supervision and Teaching:

2026 Corresponding organizer of ICRA 2026 Workshop on manipulation robustness, Vienna | *Website*

2025 Guest lecture at Duke University's graduate course, Robotic Manipulation, 2025 Spring

2024 Invited talk at Prof. Joel Burdick's group, California Institute of Technology

2024 - Now Reviewer for RA-L, ICRA, IROS, ISRR

2024 - 2026 Supervised master students, Kei Ikemura (next: KTH), Hongjia Liu (next: UCSD), Li Chen, Gawtam C. Ramesh, Zhenyuan Liang

2024, 2025 Teaching assistant in Introduction to Robotics, DD2410, KTH

Experiences

2024 **Mobile Manipulation Challenge, ABB**, *Västerås, Sweden*
Team member of KTH: Custom gripper co-design for multi-task manipulation | Slides

2021 **Robotic Systems Lab, ETH**, *Zürich*, *Graduate Researcher*, with Prof. Marco Hutter
Master thesis: Mobile door state estimation and parameter identification | Thesis
Semester thesis: Contact-implicit MPC for mobile manipulation | Video | Thesis

2020 **Computer Vision and Geometry Lab & Microsoft**, *Zürich*
Course project: Object mesh reconstruction using RGB-D cameras. | Thesis | Git | Video

2019 **SAIC Motor**, *Shanghai*
Bachelor thesis: Strategy optimization of autonomous emergency braking

2018 **Institute of Design and Control Engineering, SJTU**, *Shanghai*
3-DOF plum processing system for surface-curving and core-deprivation. | Video

2017 **Institute of Intelligent Vehicle, SJTU**, *Shanghai*
Structural design and locomotion formulation of a snake robot. | Video

Awards & Grants

2024 Travel Grant from the Karl Engvers Foundation

2021, 2022 ETH Scholarship for International Students

2019 Excellence Award, BAIC Automobile Invitational Tournament

2018 Tan Kah Kee Invention Award

2016, 2018 General Motors Scholarship

2017 Huawei Scholarship

2016, 2017 Zhiyuan College Honors Scholarship

Skills

Robot Franka Panda, Allegro hand, Ufactory Xarm 7, ANYmal legged robot, etc.

Programming Python, C++, MATLAB, C, C#

Tools ROS/ROS2, Genesis, Mujoco, Pybullet, Isaac Gym, Git, Docker, Blender, PyTorch, Codex, Simulink, Unity3D, CAD (ANSYS, SolidWorks, CATIA, NX Unigraphics)

Language English (TOEFL 109/120), Chinese (native), German (Goethe B1)